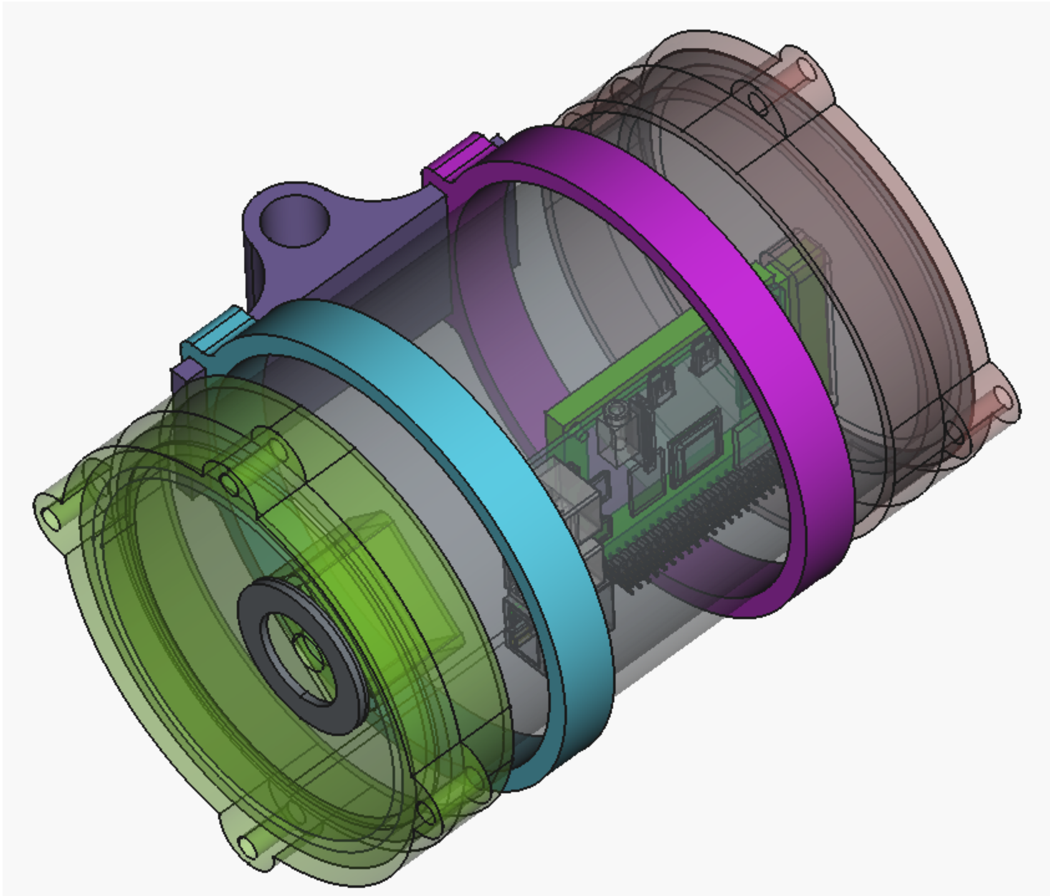


BIRDPIPE ASSEMBLY INSTRUCTIONS

Low-cost autonomous bird sound recorder based on Raspberry Pi and USB audio devices

*This document includes information of the first version of BirdPipe assembly and material costs. Document is written on **16 august 2026**, prices and components availability may vary in future.*

REQUIREMENTS

Access to a 3D printer **or** the ability to use commercially available pipe components
PETG or ABS filament for 3D printing
USB microphone
Raspberry Pi 4B (8 GB RAM or more)
USB cables
Battery pack or USB charger
110 mm waste pipe or another pipe with an outer diameter of 50 mm
Zip ties
Optional wind protection material (if the microphone does not include a windscreen)
Optional weatherproof acoustic membrane
Optional Mesh netting (for insect protection, if needed)
Optional GNSS HAT for Raspberry Pi + antennas
Optional USB 4G/LTE modem + antennas

COMPONENTS USED IN THE TEST DEVICE

3D Printer: Bambu Lab X1 Carbon
Printing material: General-purpose PETG, printed with a layer height of 0.12 mm
USB microphone: Dodotronic Ultramic 384K EVO (capable of recording ultrasound)
Raspberry Pi 4B, 8 GB RAM
GNSS module: Waveshare SX1262
Mobile modem: Brovi (Huawei) E3372-325 LTE/4G/3G USB modem
Power source: Generic 30,000 mAh power bank with USB-C connection
Housing: Generic 110 mm PVC waste pipe
Acoustic Vent: Gore Acoustic Vent AVP series (1 cm diameter)
Generic metallic mesh

1. SHELL STRUCTURE

Shell is combined by following 4 components:

- 50 mm waste pipe
- printable_parts_and_instructions/BirdPipe-110mmEndForDodotronic.stl
- printable_parts_and_instructions/BirdPipe-110mmEndForRaspHolder.stl
- printable_parts_and_instructions/BirdPipe-RaspberryPiHolder.stl

The cost of a 110 mm PVC waste pipe in Finland is approximately €6.55 for a 2 m section. Only 20 cm of pipe is required for a single BirdPipe unit, making the housing material cost very low. The pipe can be cut easily using a handsaw, jigsaw, or a dedicated pipe-cutting tool. When cutting or handling the pipe, always use appropriate protective equipment, including safety glasses and protective gloves. Remove and round any sharp edges with sand paper after cutting to prevent injury and ensure proper fit of the assembled components.

The required parts can be printed using the STL files provided above with a 3D printer capable of printing PETG or ABS materials. We recommend using white filament to reduce heat absorption from direct sunlight. Alternatively, standard pipe end caps and sealing components available from hardware stores can be used instead of the 3D-printed parts. For the BirdPipe prototype assembly described in this document, all parts were printed using a Bambu Lab X1 Carbon 3D printer.

Alternatively, standard off-the-shelf pipe end caps and fittings can be used in place of the 3D-printed parts. This eliminates the need for a 3D printer entirely and allows the enclosure to be assembled using only commercially available plumbing components. The selection of suitable fittings will depend on the pipe type and dimensions available in the local market. The availability of suitable pipe end caps may vary between countries and manufacturers. It is important to identify a commercially available end cap that fits securely onto the selected pipe after it has been cut to length. Before purchasing materials, verify the pipe dimensions and compatibility of the end caps to ensure a proper fit and adequate weather protection for the assembled device.

The end cap and holder components are designed for a press-fit assembly, allowing easy installation and removal. Under normal circumstances, no glue or adhesive is required.

2. RASPBERRYPI

Mount the RaspberryPi to the BirdPipe-RaspberryPiHolder using small zip ties passed through the mounting holes of the Raspberry Pi circuit board. In most cases, fastening two opposite corners provides adequate support and keeps the board securely positioned inside the enclosure. Ensure that the zip ties do not interfere with connectors, cables, or electronic components.

When assembling, ensure that the USB ports face upwards as the following picture shows. Insert the SD card before tightening the zip ties.

3. ADDING MICROPHONE AND CABLES

In our device, we used the Dodotronic Ultramic 384K EVO, an ultrasonic microphone capable of recording frequencies beyond the range of human hearing. The microphone provides excellent audio quality but is relatively expensive, costing approximately €280. However, the BirdPipe system is basically Linux and is not limited to these specific microphone models. Most USB microphones compatible with Raspberry Pi and Raspberry Pi OS (Raspbian) can also be used for audio recording, allowing users to choose a more affordable option depending on their requirements.

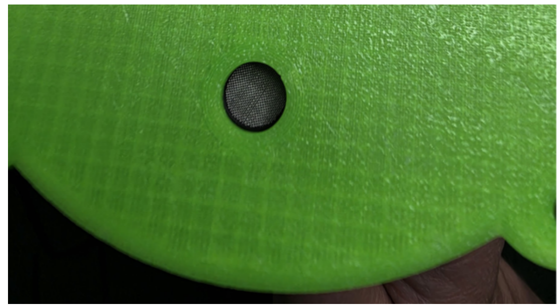
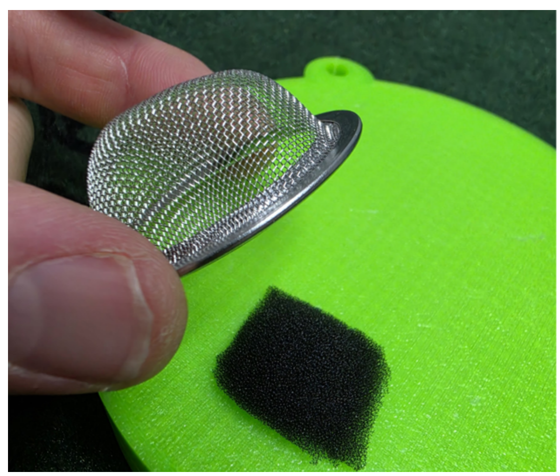
		
Soundblaster audiocard and analog microphone	Unnamed USB microphone	Dodotronic

Structure (printable_parts_and_instructions/BirdPipe-110mmEndForDodotronic.stl) holds Dodotronic, but with drill, hotglue and skill it is easy to change another microphone variant to cut remarkably costs.

Another cable is used for USB power. In our prototype, we used a 3,000 mAh power bank. Alternatively, the device can be powered from a wall outlet using a USB power adapter, or from a solar-powered system using an appropriate USB or 12 V power converter. It is important to understand the safety considerations associated with batteries, solar panels, and electrical installations. **Always follow the manufacturer's instructions and applicable local regulations. If you are unsure about any aspect of the installation, consult a qualified local specialist for guidance and safety information.**

4. MICROPHONE PROTECTION

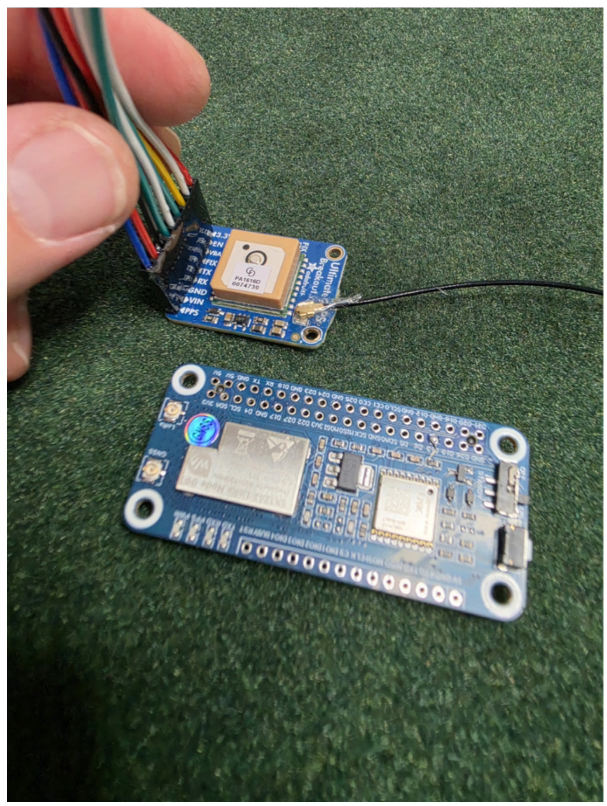
The open microphone hole is a vulnerable part of the enclosure. Care should be taken to prevent water, dust, insects, and other debris from entering the device through the microphone opening. The BirdPipe enclosure is not fully waterproof, and additional protection may be required depending on the installation environment. Your need for protection may vary depending on the research environment.

The open microphone hole is a vulnerable part of the enclosure.	
Gore Acoustic Vent AVP series (1 cm diameter) take care of rain and water spills. We also secure corners of the membrane by using general super glue.	
A small amount of the superlon foam can be used as a wind shield for the microphone. To help prevent insects from entering the device, a piece of fine mesh can be installed over the opening with MicrophoneProtector.stl. Or use existing solutions: One half of a tea strainer provides a convenient and inexpensive insect screen.	

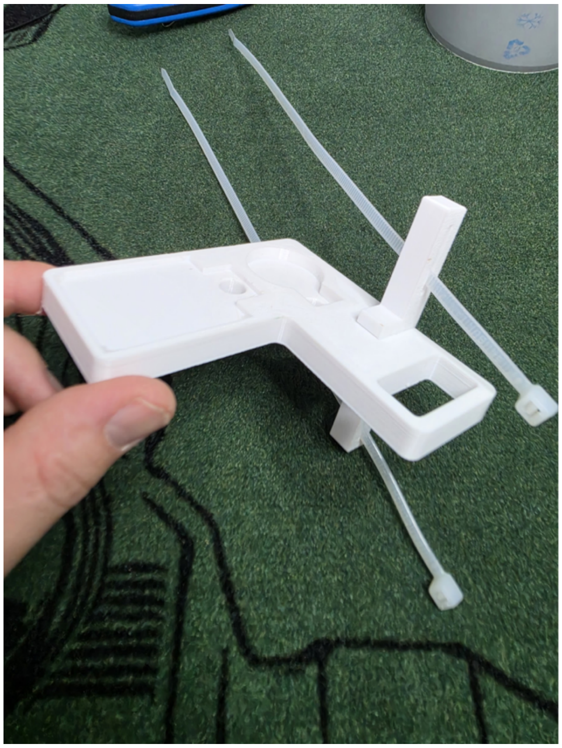
Sensitive microphones can be affected by both electromagnetic interference and acoustic resonance inside the enclosure. To reduce unwanted noise, the microphone should be isolated from vibrating surfaces. The Dodotronic microphone mount provides a tight fit that minimizes acoustic resonance. If an optional LTE modem is used, it is recommended to place it outside the enclosure and as far from the microphone as practical to reduce potential radio-frequency interference.

5. GNSS

GNSS “GPS” functionality can be added to the device using a GNSS receiver. In addition to providing accurate location data, GNSS delivers a precise time reference, even in environments where no internet connection is available. Several GNSS modules are available for the Raspberry Pi platform and can be installed as HAT expansion boards. In this project, we use the Waveshare SX1262 module. Integrating GNSS improves the temporal accuracy of audio recordings and makes it easier to locate deployed devices in the field.

	
Waveshare SX1262	Waveshare SX1262 and small GNSS alternative with integrated ceramic antenna

External antennas can be mounted at a suitable distance from the device using a structure similar to the one shown below. Cable pass-through holes for the antenna connections should be drilled in the rear end cap of the enclosure. Positioning antennas outside the enclosure and away from sensitive electronics can help improve signal quality and reduce interference.



Following files needed for antenna assembly:
external-antenna-structure-LTE-holder.stl
external-antenna-structure-Plate.stl
external-antenna-structure-Tree-Attachment.stl

6. WATERPROOF CONSIDERIATIONS

The device is not completely waterproof. The following measures are recommended to improve its resistance to water:

- Seal all external cable entries.
- Silicone sealant can be applied to the groove of the end caps; however, the pipe must be cut perfectly to ensure a proper seal.
- Additional waterproof tape can be applied on top of the device's assembly joints for extra protection.
- If there is no need to open the front end cap during operation, it can be permanently glued in place. In that case, only the rear end cap should remain removable for maintenance and access.

7. MOUNTING AND SECURITY

The device can be mounted using the holes integrated into the end caps and a rubber-coated steel cable, for example to a tree, pole, or similar structure. The cable can be locked, which helps prevent unauthorized access to the device and reduces the risk of component theft.